



NCI Protocol #: 10323

Patient ID:

Date:

1 of 13

Patient Name:

Date Collected:

Biopsy Site: **Bone Marrow**

Sample ID:

Telephone:

Tumor Content (%): **90**

Patient DOB:

Referring Physician:

Primary Diagnosis: **Acute myeloid leukemia**

MoCha ID:

Fax:

Preanalytics to include tumor enrichment as required is performed by Roswell Park Comprehensive Cancer Center; Buffalo; NY 14203

Relevant Acute Myeloid Leukemia Findings

Gene	Finding	Gene	Finding
ABL1	None detected	MECOM	None detected
ASXL1	None detected	MLLT3	KMT2A-MLLT3 fusion
CEBPA	None detected	MYH11	None detected
CREBBP	None detected	NPM1	None detected
FLT3	FLT3 D835E (FLT3 TKD mutation)	NUP214	None detected
IDH1	None detected	RARA	None detected
IDH2	None detected	RUNX1	None detected
KMT2A	KMT2A-MLLT3 fusion	TP53	None detected

Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IA	FLT3 D835E FLT3 TKD mutation	gilteritinib ¹ midostaurin + chemotherapy ¹ allogeneic stem cells azacitidine decitabine venetoclax + chemotherapy	None	21

Public data sources included in relevant therapies: FDA¹, NCCN

Public data sources included in prognostic and diagnostic significance: NCCN

Tier Reference: Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

Relevant Biomarkers (continued)

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IA	KMT2A-MLLT3 fusion	allogeneic stem cells	None	14
		azacitidine		
		cytarabine		
		cytarabine + daunorubicin +		
		etoposide		
		cytarabine + etoposide + idarubicin		
		cytarabine + fludarabine + idarubicin		
		+ filgrastim		
		daunorubicin		
		decitabine		
		gemtuzumab ozogamicin +		
		chemotherapy		
		idarubicin		
		liposomal cytarabine-daunorubicin		
		CPX-351		
		mitoxantrone		
		venetoclax + chemotherapy		
Prognostic significance: ELN 2022: Intermediate				

Public data sources included in relevant therapies: FDA1, NCCN
Public data sources included in prognostic and diagnostic significance: NCCN
Tier Reference: Li et al. Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists. J Mol Diagn. 2017 Jan;19(1):4-23.

Variant Details

DNA Sequence Variants

Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Coverage	Transcript	Variant Effect
FLT3	p.(D835E)	c.2505T>G	COSM788	chr13:28592640	42.38%	1968	NM_004119.3	missense

Gene Fusions (RNA)

Genes	Variant ID	Locus
KMT2A-MLLT3	KMT2A-MLLT3.K8M6	chr11:118353210 - chr9:20365742

Low Coverage Regions:

NA

Comments:

NA

Report Signed By

Name	Role	Date
Shananawaz Jiwani	Pathologist	06:07:25 AM

Biomarker Descriptions

FLT3 (fms related receptor tyrosine kinase 3)

Background: The FLT3 gene encodes the fms related tyrosine kinase 3, a tyrosine kinase receptor that is a member of the class III receptor tyrosine kinase family that also includes PDGFR, FMS, and KIT¹. FLT3 is highly expressed in hematopoietic progenitor cells². Genomic alterations in FLT3 activate downstream oncogenic pathways including PI3K/AKT/mTOR and RAS/RAF/MEK/ERK pathways which promote cellular proliferation, survival, and inhibition of differentiation¹.

Alterations and prevalence: Somatic mutations occur in approximately 30% of acute myeloid leukemia (AML), 7-10% of melanoma, and up to 8% of uterine cancer^{3,4,5,6}. The most common activating FLT3 mutations are internal tandem duplications (ITD) that range from 3 to 400 base pairs in length within exons 14 and 15 in the juxtamembrane (JM) domain⁷. The second most frequent mutations are point mutations in exon 20 within the tyrosine kinase domain (TKD)⁸. FLT3 is amplified in up to 8% of colorectal cancer, 3% of stomach cancer, and is commonly overexpressed in AML^{5,6,9}.

Potential relevance: FLT3 rearrangements are recognized by the World Health Organization (WHO) as one of the possible molecular abnormality requirements that define myeloid/lymphoid neoplasms with eosinophilia and tyrosine kinase gene fusions¹⁰. The presence of FLT3-ITD confers poor prognosis in myelodysplastic syndrome (MDS)¹¹. Concurrent expression of FLT-ITD with mutant or wild-type NPM1 (when lacking adverse risk genetic lesions) confers intermediate risk in AML^{12,13}. FLT3 TKD mutation at D835 confers poor prognosis in MDS¹¹. Midostaurin¹⁴ (2017) and gilteritinib¹⁵ (2018) are kinase inhibitors approved for AML patients with FLT3-ITD and TKD mutations including D835 and I836 mutations. The FDA granted fast track designations in 2017 to crenolanib¹⁶ and in 2022 to tuspentinib (HM43239)¹⁷ for FLT3 mutation-positive relapsed or refractory AML. In 2018 the FDA granted breakthrough therapy designation to quizartinib¹⁸ for AML with FLT3-ITD. A phase II trial testing crenolanib in 34 patients with FLT3-ITD and TKD mutated relapsed/refractory AML, reported that FLT3 inhibitor naïve patients demonstrated a longer overall survival (OS) and event free survival (EFS) in comparison to previously treated patients (median OS: 55 weeks vs 13 weeks; median EFS: 13 weeks vs 7 weeks)¹⁹. Another phase II trial of crenolanib with chemotherapy in newly diagnosed FLT3 mutated AML reported complete remission in 24/29 (83%) patients²⁰. Several multi-targeted tyrosine kinase inhibitors such as sorafenib (2005), sunitinib (2006), cabozantinib (2012), and ponatinib (2012) are FDA approved and include FLT3 as a target. Sorafenib is recommended in combination with chemotherapy in FLT3-ITD mutated AML¹².

KMT2A (lysine methyltransferase 2A)

Background: The KMT2A gene encodes the lysine methyltransferase 2A protein, a transcriptional coactivator and histone H3 lysine 4 (H3K4) methyltransferase. KMT2A, also known as mixed lineage leukemia (MLL), is part of the SET domain protein methyltransferase superfamily. KMT2A influences epigenetic regulation by means of its methyltransferase activity, which regulates a variety of cellular functions including neurogenesis, hematopoiesis, and osteogenesis²¹. Located at the chromosomal position 11q23, KMT2A is the target of recurrent chromosomal rearrangements observed in several leukemia subtypes including MLL, acute myeloid leukemia (AML), and acute lymphoblastic leukemia (ALL)²². Such translocations encode KMT2A fusion proteins that are oncogenic with simultaneous loss of KMT2A H3K4 methyltransferase activity²². Loss of methyltransferase activity along with partner gene gain of function contributes to increased HOX gene expression and promotes the transformation of hematopoietic cells into leukemic stem cells^{22,23,24,25}.

Alterations and prevalence: KMT2A fusions are observed in 3-10% of AML cases with the highest frequencies in therapy-related AML (9%) and patients younger than 60 years (5%)^{12,22,26}. KMT2A rearrangements including t(4;11)(q21;q23)/AFF1-KMT2A, t(9;11)(p22;q23)/MLLT3-KMT2A, t(11;19)(q23;p13.3)/KMT2A-MLLT1, t(10;11)(p12;q23)/MLLT10-KMT2A, and t(6;11)(q27;q23)/AFDN-KMT2A translocations account for about 80% of all KMT2A rearranged leukemias²². In infant acute leukemic cases, KMT2A rearrangement is reported in up to 70% of those diagnosed with either AML or ALL^{22,27,28}. Mutations in KMT2A are also reported in diverse solid tumors including 10-20% of melanoma, stomach, bladder, and uterine cancers and around 5% of lung and head and neck cancers⁶.

Biomarker Descriptions (continued)

KMT2A alterations observed in solid tumors include nonsense or frameshift mutations which result in KMT2A truncation and loss of methyltransferase activity^{6,29}.

Potential relevance: KMT2A fusions are associated with variable prognosis based on the partner genes involved in the fusion^{12,13}. For example, t(6;11)(q27;q23)/AFDN-KMT2A fusions are associated with poor prognosis whereas, t(9;11)(p22;q23)/MLLT3-KMT2A fusions confer more favorable or intermediate prognosis in AML^{30,31,32}. Additionally, 11q23 rearrangements define an unfavorable karyotype in patients diagnosed with primary myelofibrosis (PMF) and may confer intermediate to high risk depending on concurrent cytogenetic abnormalities³³. KMT2A fusion is also associated with poor risk in ALL³⁴. In 2022, the FDA granted breakthrough therapy designation to the oral menin inhibitor, revumenib³⁵, for the treatment of patients with relapsed or refractory acute leukemia harboring a KMT2A rearrangement.

MLLT3 (MLLT3 super elongation complex subunit)

Background: The MLLT3 gene encodes the AF-9 protein, a component of the super elongation complex (SEC) responsible for rapid transcriptional induction of immediate response and developmental genes³⁶. MLLT3 possesses a conserved YEATS domain that is involved in the regulation of chromatin structure and gene transcription³⁷. Specifically, the YEATS domain binds acylated histone H3 with preferential recognition of H3 that has been crotonylated— a post-translational modification enriched on active gene promoters that confers resistance to transcriptional repressors^{37,38,39,40}. MLLT3 is the target of recurrent chromosomal translocations in hematological malignancies including acute myeloid leukemia (AML)^{41,42}.

Alterations and prevalence: Chromosomal rearrangements involving MLLT3 has been observed with various fusion partners. The most common rearrangement involves the t(9;11)(p21;q23) translocation, which results in the MLLT3-KMT2A fusion⁴³. In AML, the t(9;11)(p21;q23) translocation is observed in 5-12% of pediatric and 1-2% of adult AML cases^{43,44}.

Potential relevance: The t(9;11)(p21;q23) rearrangement, resulting in MLLT3-KMT2A fusion, confers intermediate risk in AML^{12,13}.

Relevant Therapy Summary

In this cancer type

In other cancer type

In this cancer type and other cancer types

No evidence

FLT3 D835E			
Relevant Therapy	FDA	NCCN	Clinical Trials*
gilteritinib			
midostaurin + cytarabine + daunorubicin			
Allogeneic hematopoietic stem cell transplantation			
azacitidine			
decitabine			
midostaurin + cytarabine			
midostaurin + daunorubicin			
venetoclax + azacitidine			
venetoclax + cytarabine			

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

FLT3 D835E (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
venetoclax + decitabine	×	●	×
chemotherapy, midostaurin	×	×	● (III)
crenolanib, chemotherapy	×	×	● (III)
crenolanib, midostaurin, chemotherapy	×	×	● (III)
gemtuzumab ozogamicin, chemotherapy, gilteritinib	×	×	● (III)
gilteritinib, midostaurin, chemotherapy	×	×	● (III)
SKLB1028	×	×	● (III)
chemotherapy, venetoclax	×	×	● (II/III)
chemotherapy, gilteritinib, midostaurin	×	×	● (II)
natural killer cell therapy, stem cell therapy	×	×	● (II)
DSP-5336	×	×	● (I/II)
gilteritinib + chemotherapy, gilteritinib + venetoclax + chemotherapy	×	×	● (I/II)
gilteritinib, chemotherapy	×	×	● (I/II)
gilteritinib, venetoclax, chemotherapy	×	×	● (I/II)
KDS-1001, cytokine, tacrolimus, mycophenolate mofetil, radiation therapy, chemotherapy, allogeneic stem cells	×	×	● (I/II)
midostaurin, gemtuzumab ozogamicin, chemotherapy	×	×	● (I/II)
NMS-P948	×	×	● (I/II)
venetoclax, gilteritinib, chemotherapy	×	×	● (I/II)
gemtuzumab ozogamicin, midostaurin, chemotherapy	×	×	● (I)
iadademstat, gilteritinib	×	×	● (I)
PHI-101	×	×	● (I)

KMT2A-MLLT3 fusion

Relevant Therapy	FDA	NCCN	Clinical Trials*
Allogeneic hematopoietic stem cell transplantation	×	●	×

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

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Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2023.06(006).

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

KMT2A-MLLT3 fusion (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
azacitidine	×	●	×
cytarabine	×	●	×
cytarabine + daunorubicin + etoposide	×	●	×
cytarabine + etoposide + idarubicin	×	●	×
cytarabine + fludarabine + idarubicin + filgrastim	×	●	×
daunorubicin	×	●	×
decitabine	×	●	×
gemtuzumab ozogamicin + cytarabine + daunorubicin	×	●	×
gemtuzumab ozogamicin + cytarabine + fludarabine + idarubicin + filgrastim	×	●	×
gemtuzumab ozogamicin + daunorubicin	×	●	×
idarubicin	×	●	×
liposomal cytarabine-daunorubicin CPX-351	×	●	×
mitoxantrone	×	●	×
venetoclax + azacitidine	×	●	×
venetoclax + cytarabine	×	●	×
venetoclax + cytarabine + fludarabine + idarubicin + filgrastim	×	●	×
venetoclax + decitabine	×	●	×
AVC-WT1 Therapy	×	×	● (II)
chemotherapy	×	×	● (II)
natural killer cell therapy, stem cell therapy	×	×	● (II)
DSP-5336	×	×	● (I/II)
KDS-1001, cytokine, tacrolimus, mycophenolate mofetil, radiation therapy, chemotherapy, allogeneic stem cells	×	×	● (I/II)
revumenib	×	×	● (I/II)
revumenib + venetoclax + chemotherapy	×	×	● (I/II)
ziftomenib	×	×	● (I/II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

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Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

KMT2A-MLLT3 fusion (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
BMF-219	×	×	● (I)
nintedanib, chemotherapy	×	×	● (I)
pemigatinib, chemotherapy	×	×	● (I)
revumenib, chemotherapy, steroid	×	×	● (I)
venetoclax, chemotherapy	×	×	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Clinical Trials Summary

FLT3 D835E + KMT2A-MLLT3 fusion

NCT ID	Title	Phase
NCT04988555	A Phase I/II, Open-Label, Dose-Escalation, Dose-Expansion Study of DSP-5336 in Adult Acute Leukemia Patients With and Without Mixed Lineage Leukemia (MLL)-Rearrangement or Nucleophosmin 1 (NPM1) Mutation.	I/II

FLT3 D835E

NCT ID	Title	Phase
NCT03250338	Phase III Randomized, Double-blind, Placebo-controlled Study Investigating the Efficacy of the Addition of Crenolanib to Salvage Chemotherapy Versus Salvage Chemotherapy Alone in Subjects < or = 75 Years of Age With Relapsed/Refractory FLT3 Mutated Acute Myeloid Leukemia	III
NCT03258931	Phase III Randomized Study of Crenolanib Versus Midostaurin Administered Following Induction Chemotherapy and Consolidation Therapy in Newly Diagnosed Subjects With FLT3 Mutated Acute Myeloid Leukemia	III
NCT04716114	A Phase III, Open-Label, Multicenter, Randomized Study of SKLB1028 Versus Salvage Chemotherapy in Patients With FLT3-mutated Acute Myeloid Leukemia Refractory to or Relapsed After First-line Treatment(ALIVE)	III
NCT05520567	A Phase I/II, Multicenter, Open-Label, Randomized Dose Ranging and Expansion Study of the Combination of Gilteritinib, Venetoclax and Azacitidine in Patients With Newly Diagnosed FLT3 Mutated Acute Myeloid Leukemia (AML) Not Eligible for Intensive Induction Chemotherapy	I/II
NCT05010122	A Phase I/II Study of ASTX727, Venetoclax, and Gilteritinib for Patients With Acute Myeloid Leukemia or High-Risk Myelodysplastic Syndrome With an Activating FLT3 Mutation	I/II

Clinical Trials Summary (continued)

FLT3 D835E (continued)

NCT ID	Title	Phase
NCT05546580	An Escalation/Expansion, Open Label, Multicenter Study of Iadademstat and Gilteritinib in Patients With Relapsed or Refractory Acute Myeloid Leukemia (R/R AML) With FMS-like Tyrosine Kinase Mutation (FLT3 Mut+): The FRIDA Study	I
NCT04174612	A Phase III, Prospective, Randomized Multi-center Intervention Trial of Early Intensification in AML Patients Bearing FLT3 Mutations Based on Peripheral Blast Clearance: A MYNERVA-GIMEMA Study	III
NCT04293562	A Phase III Randomized Trial for Patients With De Novo AML Comparing Standard Therapy Including Gemtuzumab Ozogamicin (GO) to CPX-351 With GO, and the Addition of the FLT3 Inhibitor Gilteritinib for Patients With FLT3 Mutations	III
NCT04027309	A Phase III, Multicenter, Open-label, Randomized, Study of Gilteritinib Versus Midostaurin in Combination With Induction and Consolidation Therapy Followed by One-year Maintenance in Patients With Newly Diagnosed Acute Myeloid Leukemia (AML) or Myelodysplastic Syndromes With Excess Blasts-2 (MDS-EB2) With FLT3 Mutations Eligible for Intensive Chemotherapy (HOVON 156 AML / AMLSG 28-18)	III
NCT02416388	Phase II/III Randomized Study to Improve Overall Survival in 18 to 60 Year-old Patients, Comparing Daunorubicin Versus High Dose Idarubicin Induction Regimens, High Dose Versus Intermediate Dose Cytarabine Consolidation Regimens, and Standard Versus Mycophenolate Mofetil Prophylaxis of Graft Versus Host Disease in Allografted Patients in First CR : a Backbone InterGroup-1 Trial	II/III
NCT02115295	Phase II Study of Cladribine Plus Idarubicin Plus Cytarabine (ARAC) in Patients With AML, HR MDS, or Myeloid Blast Phase of CML	II
NCT04240002	A Phase I/II, Multicenter, Open-Label, Single Arm, Dose Escalation and Expansion Study of Gilteritinib (ASP2215) Combined With Chemotherapy in Children, Adolescents and Young Adults With FMS-like Tyrosine Kinase 3 (FLT3)/Internal Tandem Duplication (ITD) Positive Relapsed or Refractory Acute Myeloid Leukemia (AML)	I/II
NCT04385290	Midostaurin + Gemtuzumab OzogAmlcin Combination in First-line Standard Therapy for Acute Myeloid Leukemia (MOSAIC).	I/II
NCT03922100	A Phase I/II Study of NMS-03592088, a FLT3, KIT and CSF1R Inhibitor, in Patients With Relapsed or Refractory AML or CMML	I/II
NCT03900949	A Phase I Study to Evaluate the Safety and Tolerability of Gemtuzumab Ozogamicin and Midostaurin When Used in Combination With Standard Cytarabine and Daunorubicin Induction for Newly Diagnosed FLT3-mutated Acute Myeloid Leukemia	I
NCT05024552	A Phase I Study of Vyxeos Plus Gilteritinib in Relapsed or Refractory, FLT3-Mutated Acute Myeloid Leukemia	I
NCT04842370	A Prospective, Phase Ia/Ib, Multicenter Study to Evaluate the Safety, Tolerability, Pharmacokinetics and Pharmacodynamics of the FLT3 Inhibitor, PHI 101, Alone in Subjects With Relapsed or Refractory Acute Myeloid Leukemia (AML).	I
NCT02727803	Personalized NK Cell Therapy in Cord Blood Transplantation	II
NCT03013998	A Master Protocol for Biomarker-Based Treatment of AML (The Beat AML Trial)	I/II

Clinical Trials Summary (continued)

FLT3 D835E (continued)

NCT ID	Title	Phase
NCT05115630	A Phase I/II Clinical Trial of "Off-the-shelf" NK Cell Administration in Combination With Allogeneic SCT to Decrease Disease Relapse in Patients With High-risk Myeloid Malignancies Undergoing Matched Related, Matched Unrelated, One Antigen Mismatched Unrelated, or Haploidentical Stem-cell Transplantation	I/II

KMT2A-MLLT3 fusion

NCT ID	Title	Phase
No NCT ID	A Phase IIa Study Of a AVC-WT1 Therapy To Treat Acute Myeloid Leukemia And High-Risk Myelodysplastic Syndromes With MRD-Positive Complete Remission Or Partial Remission	II
No NCT ID	International Multicenter, Open-Label, Phase II Study to Treat Molecular Relapse of Pediatric Acute Myeloid Leukemia With Azacitidine	II
NCT04659616	A Multicenter, Phase I, Open-Label Study of the FGFR Inhibitor Pemigatinib (INCB054828) Administered After Chemotherapy in Newly Diagnosed Acute Myeloid Leukemia (AML) With Adverse or Intermediate Risk Cytogenetics	I
NCT02727803	Personalized NK Cell Therapy in Cord Blood Transplantation	II
NCT05115630	A Phase I/II Clinical Trial of "Off-the-shelf" NK Cell Administration in Combination With Allogeneic SCT to Decrease Disease Relapse in Patients With High-risk Myeloid Malignancies Undergoing Matched Related, Matched Unrelated, One Antigen Mismatched Unrelated, or Haploidentical Stem-cell Transplantation	I/II
NCT04065399	AUGMENT-101: A Phase I/II, Open-label, Dose-Escalation and Dose-Expansion Cohort Study of SNDX 5613 in Patients With Relapsed/Refractory Leukemias, Including Those Harboring an MLL/KMT2A Gene Rearrangement or Nucleophosmin 1 (NPM1) Mutation	I/II
NCT03013998	A Master Protocol for Biomarker-Based Treatment of AML (The Beat AML Trial)	I/II
NCT04067336	A Phase I/II First in Human Study of the Menin-MLL(KMT2A) Inhibitor KO-539 in Patients With Relapsed or Refractory Acute Myeloid Leukemia	I/II
NCT05153330	A Phase I First-in-human Dose-escalation and Dose-expansion Study of BMF-219, an Oral Irreversible Menin Inhibitor, in Adult Patients With Acute Leukemia (AL), Diffuse Large B-cell Lymphoma (DLBCL), and Multiple Myeloma (MM).	I
NCT03513484	Phase Ib, Open Label, Combination Study of Nintedanib With 5-Azacitidine in Acute Myeloid Leukemia Characterized by HOX Gene Overexpression, That Are Not Candidates of Intensive Chemotherapy	I
NCT05326516	AUGMENT-102: A Phase I, Open-label, Dose-escalation Study to Evaluate Safety, Tolerability and Preliminary Anti-Leukemic Activity of SNDX-5613 in Combination With Chemotherapy in Patients With Relapsed/Refractory Leukemias Harboring a KMT2A/MLL Gene Rearrangement or Nucleophosmin 1 Mutation (mNPM1)	I
NCT03613532	A Phase I Study of Adding Venetoclax to a Reduced Intensity Conditioning Regimen and to Maintenance in Combination With a Hypomethylating Agent After Allogeneic Hematopoietic Cell Transplantation for Patients With High Risk AML, MDS, and MDS/MPN Overlap Syndromes	I

Clinical Trials Summary (continued)

KMT2A-MLLT3 fusion (continued)

NCT ID	Title	Phase
NCT03844815	Phase I Study of Venetoclax in Combination With Decitabine 10-Day Regimen in Subjects With Acute Myeloid Leukemia	I

Assay Information

Methodology, Scope, and Application:The NCI Myeloid Assay v2 (NMAv2) is a next-generation sequencing (NGS) assay which identifies pre-defined and novel genomic variants covering 45 DNA genes and 35 fusion drivers that are categorized into 3 mutation types: single nucleotide variants (SNVs), insertions/deletions (Indels) including FLT3 internal tandem duplications (ITDs), and gene fusions. The assay utilizes Thermo Fisher Scientific’s Oncomine® Myeloid Assay GX v2, a NGS assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from blood and bone marrow mononuclear cells for sequencing on the Ion Torrent Genexus Integrated Sequencer and analyzed by the Ion Torrent Genexus pipeline version 6.6.2.1. The NMAv2 can reliably identify the presence or absence of 1661 hotspot variants and 779 reported fusions compared to the Human Reference Genome hg19, including the genes listed below. The NMAv2 is a laboratory developed test designed to detect gene mutations for major myeloid disorders.

Analytical Sensitivity and Specificity:The NMAv2 has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 98.97% sensitivity, 100% specificity and an overall reproducibility with a mean positive percent agreement of 99% and a mean negative percent agreement of 100% during validation of the assay.

Hereditary and Germline Mutations:This assay examines cancer cells only and does not examine normal (non-cancer) cells. Mutations detected by the assay may be present only in the cancer cells, or in every cell of the body (including non-cancer cells). This test also cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient’s history includes any one or more features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient’s), it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal (i.e. non-cancer) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

Report Content:Thermo Fisher Scientific’s Ion Torrent Oncomine Knowledgebase Reporter (OKR) software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of OKR. The data presented here is from a curated knowledgebase of publicly available information but may not be exhaustive. FDA information was sourced from www.fda.gov and is current as of this version. NCCN information was sourced from www.nccn.org and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding particular trials, search www.clinicaltrials.gov by NCT ID or search local clinical trials authority website by local identifier listed in ‘Other Identifiers.’ Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al. 2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

Genes assayed for hotspot mutations:
ABL1, ANKRD26, BRAF, CBL, CSF3R, DDX41, DNMT3A, FLT3, GATA2, HRAS, IDH1, IDH2, JAK2, KIT, KRAS, MPL, MYD88, NPM1, NRAS, PPM1D, PTPN11, SETBP1, SF3B1, SMC1A, SMC3, SRSF2, U2AF1, WT1

Genes assayed with full exon coverage:
ASXL1, BCOR, CALR, CEBPA, ETV6, EZH2, IKZF1, NF1, PHF6, PRPF8, RB1, RUNX1, SH2B3, STAG2, TET2, TP53, ZRSR2

Genes assayed for detection of fusions:

ABL1, ABL2, BCL2, BRAF, CCND1, CREBBP, EGFR, ETV6, FGFR1, FGFR2, FUS, HMGA2, JAK2, KAT6A, KAT6B, KMT2A, KMT2A-PTD (Partial Tandem Duplication), MECOM, MET, MLLT10, MRTFA (MKL1), MYBL1, MYH11, NTRK2, NTRK3, NUP214, NUP98, PAX5, PDGFRA, PDGFRB, RARA, RUNX1, TCF3, TFE3, ZNF384

Limitations of the assay

1. Mutations in homopolymer regions of >5 nucleotides may not be accurately assessed by this assay.
2. FLT3-ITDs greater than 120bps may not be detected by this assay. Sensitivity for larger FLT3-ITD's may be reduced and variant allele frequencies may be reported lower than expected.
3. Following genetic alterations will not be reported by this assay: RUNX1 p.Ser445AlafsTer149, ASXL1 p.G646Wfs*12, PRPF8 p.Arg156GlufsTer28

DISCLAIMER

This assay reports SNV, indel and FLT3-ITD mutations $\geq 5\%$ Variant Allele Frequency (VAF). This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

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